LODS-MTI: A Link-Adaptive, Orthogonal, and De-slotted Protocol for Robust and Fast RFID Missing Tag Identification

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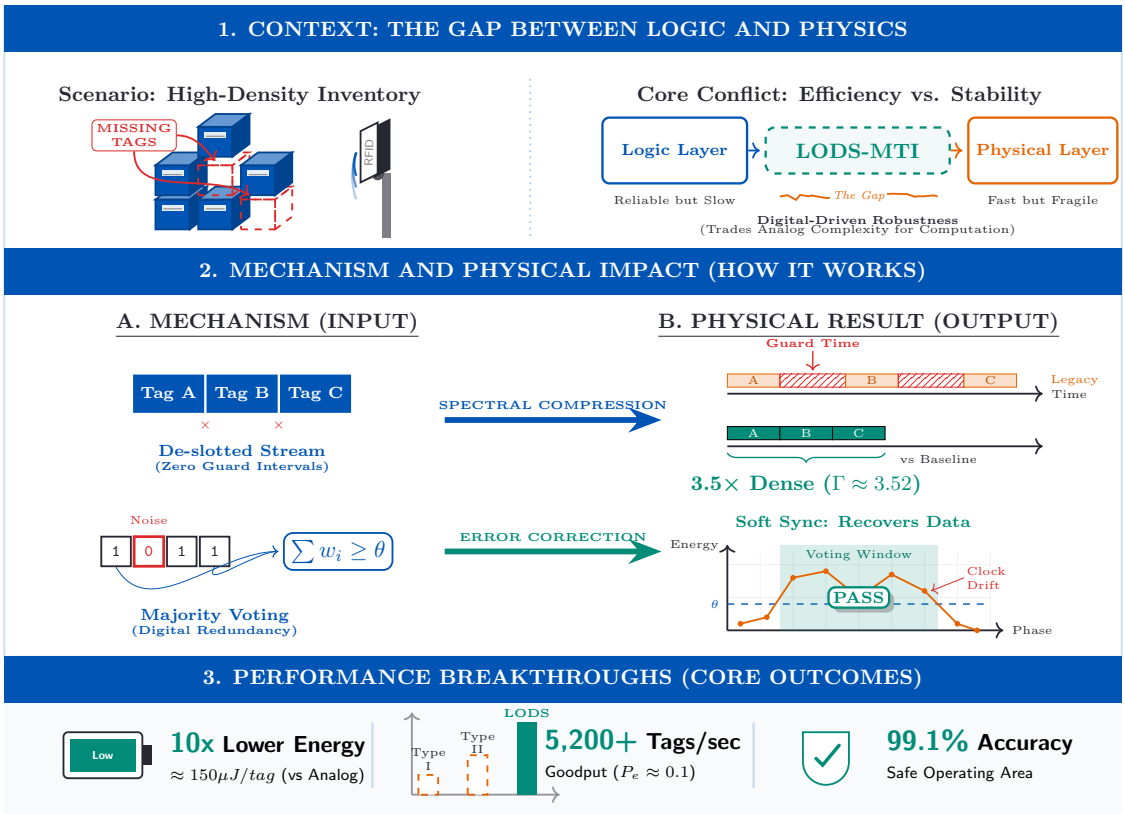
Abstract

Existing Missing Tag Identification (MTI) protocols trade logic-layer scheduling reliability for physical-layer spectral efficiency. This paper presents LODS-MTI, a Link-adaptive Orthogonal De-slotted protocol that eliminates inter-tag guard times using batch-synchronized continuous bitstreams. To mitigate synchronization loss in de-slotted transmissions, a digital majority voting mechanism corrects single-bit timing slips and accommodates 0.4% clock drift. The design uses short-term coherence through "Power-of-2" batching to replace arithmetic division with bitwise operations. A tolerance-driven feedback loop ($\epsilon = 0.30$) adjusts redundancy based on link quality to maintain a Safe Operating Area (SOA) with 99.10% accuracy within 128-bit windows. Experimental results show that LODS-MTI achieves a normalized time efficiency of $\Gamma \approx 3.52$ (a 50% increase over baselines) and sustains 5,200–8,811 tags/s goodput ($P_e \approx 0.1$) at 150 $\mu J/tag$, reducing energy consumption by approximately 10× compared to analog collision resolution schemes. Code and results are available at GitHub.

Keywords: *RFID, Missing Tag Identification, De-slotted Architecture, Perfect Hashing, Link-adaptive.*



Graphical Abstract: This graphical abstract illustrates the architecture and performance of LODS-MTI, a protocol designed to reconcile logical reliability with physical spectral efficiency in RFID systems. 1. Context: Resolves the structural conflict between stable logic-layer scheduling (Type I) and fast physical-layer collision resolution (Type II). 2. Mechanism: Replaces guard intervals with a de-slotted continuous stream for spectral compression. A Digital Majority Voting logic ($\sum w_i \geq \theta$) recovers data from synchronization drift, achieving $\Gamma \approx 3.52$. 3. Outcome: Delivers 10x lower energy ($\sim 150 \mu J/tag$) and sustains 5,200+ tags/s goodput with 99.1% accuracy.

1 Introduction

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```
\graphicalabstract{
    \includegraphics[width=0.8\linewidth]{abstract.pdf}
    \captionof{figure}{...}
}
```

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Table 1 demonstrates the default style.

Table 1. Example Table Style

Feature	Command	Note
Lead Author	`\leadauthor`	Header usage
Institution	`\institution`	Footer usage
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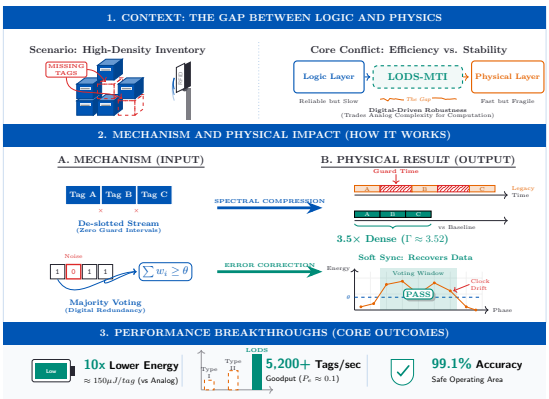

Figure 1. Example of a figure. Note that the caption is left-aligned and colored in Academic Navy.

3 Conclusion

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